

ALI AKBAR

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PROFESSIONAL SUMMARY

Electrical and Electronic Engineering graduate with a strong research background in machine learning, computer vision, multimodal AI, and biomedical data analysis. Co-author of an ACM UbiComp publication, runner-up in the UbiComp 2021 Nurse Care Activity Recognition Challenge, and contributor to an ACL Findings 2026 paper.

EDUCATION

University of Dhaka ([Website](#))

Jan 2019 - Apr 2024

Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering

CGPA: 3.52/4.00

TECHNICAL SKILLS

Languages	C, Java, Python, MATLAB, MySQL, L ^A T _E X.
ML/DL Frameworks	PyTorch, TensorFlow, scikit-learn, OpenCV, MLflow, DVC, FastAPI.
LLM & Agentic AI	LangChain, LangGraph, LangSmith, Pydantic, vector DB, Hugging Face, LLM APIs, RAG, LoRA, QLoRA.
Data Science Tools	Pandas, NumPy, Matplotlib, MS Excel.
Other Tools	Git, Docker, Streamlit.

RESEARCH & PROFESSIONAL EXPERIENCE

Intern | icddr,b

May 2026 – Present

- Developing a Linux-based shotgun metagenomics pipeline using SRA Toolkit, KneadData, and MetaPhlAn to acquire, preprocess, and generate taxonomic abundance profiles from publicly available healthy-control gut microbiome sequencing data. Integrating these data with icddr,b's existing cohort of 67 IBD patient samples to support comparative microbiome analysis and downstream machine learning studies for identifying disease-associated microbial signatures.

Research Assistant | Data & Design Lab,

Department of Computer Science and Engineering, University of Dhaka.

Feb 2025 – Dec 2025

- Collected real-time field data from substations, developed data analytics solutions, preprocessed and transformed electricity-sector datasets into structured formats for system integration and data-driven decision-making, and contributed to the development of *Biddyt Bandhu*, a domain-specific chatbot.

RESEARCH PROJECTS

Nurse Care Activity Recognition from Accelerometer Sensor Data Using Fourier- and Wavelet-based Features. May, 2021, | [paper link](#)

- Engineered and selected time-frequency features from accelerometer data using FFT, DWT, and statistical methods, achieving 87% accuracy with LightGBM for multi-class nurse activity recognition and contributing to a *runner-up finish* in the ACM UbiComp 2021 Nurse Care Activity Recognition Challenge.

Integrated Detection and Segmentation of Bacterial Cells Using a Vision Transformer-based Framework. (*Manuscript in preparation, [manuscript link](#)*)

- Developed a MaxViT-based Mask R-CNN for bacterial cell instance segmentation, outperforming ResNet50 Mask R-CNN, Mask2Former, and YOLOv11 with 79.31% mAP and 70.01% F1-score, while incorporating ROI Activation CAM for model interpretability .

Integrating Gut Microbiome Taxonomic Profiles with Supervised Machine Learning for IBS (Irritable Bowel Syndrome) Diagnosis.

- Built a gut microbiome classification pipeline using 1,109 samples, including newly sequenced Bangladeshi data, achieving 79.73% accuracy and 0.5876 MCC with optimized Logistic Regression for IBS detection.

Thinking Like a Botanist: Challenging Multimodal Language Models with Intent Driven Chain-of-Inquiry

- Contributed to the large-scale data collection for plant disease diagnosis dataset, PlantInquiryVQA, consisting of 24,964 images and 138,078 reasoning-based QA pairs for evaluating state-of-the-art multimodal language models.

(*Published in ACL Findings 2026—Acknowledged contributor for large-scale data collection. [paper link](#)*)

PROJECTS

OCR-Based Handwritten Scanned Document Extraction, Summarization, and Retrieval System.
(Live Demo)

- Developed a RAG-based system for handwritten documents using OCR, semantic retrieval, and FAISS, with source-grounded QA and a HITL workflow for refining automated case summaries.

Conversational AI Chatbot with Memory Using LangGraph and Streamlit.
(Live Demo)

- Developed a multi-turn conversational AI chatbot using LangGraph with graph-based state management, persistent memory, tool calling, and real-time streaming via Groq for context-aware interactions.

Retrieval-Augmented Generation (RAG) System for Multi-Source Website & PDF Question Answering.
(Live Demo)

- Built a web and document-based RAG pipeline that ingests multiple URLs and PDF files, performs semantic retrieval using HuggingFace embeddings and FAISS, and generates source-grounded answers with performance monitoring via LangSmith.

PUBLICATIONS

Kowshik, M. Ashikuzzaman, Yeasin Arafat Pritom, Md Sohanur Rahman, **Ali Akbar**, and Md Atiqur Rahman Ahad. **"Nurse Care Activity Recognition from Accelerometer Sensor Data Using Fourier-and Wavelet-based Features"** *In Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers*, pp. 434-439. 2021.

ACHIEVEMENTS & AWARDS

- *Runner-up*, Third Nurse Care Activity Recognition Challenge, HASCA Workshop, ACM UbiComp 2021. ([certificate](#))

LEADERSHIP & ACTIVITIES

President

Dec 2022 - Feb 2024

Luminator, Dr. Muhammad Shahidullah Hall Students' Association.

Joint General Secretary

Apr 2023 - Oct 2024

Idrakpur Bus Committee, University of Dhaka.

Organizing Secretary

May 2023 - Aug 2024

Students' Association of Munshiganj-Bikrampur, University of Dhaka.